The V_2-lemma, reduced

Exact Prüfer step formula, the contrast obstruction, and the strictly smaller profile lemma V_3'

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Abstract

Round 374 attacks the named remainder V_2 of [2, 3] by the r372 Prüfer-to-run dictionary (a theorem: sign-runs of v_2 equal $\lfloor\theta/\pi\rfloor$-dwells of the degree-accumulated Jacobi–Prüfer phase). V_2 itself is neither proved nor refuted in the construction. What is proved: an exact two-point Wronskian formula for the discrete phase step of the scaled pair (v_{N-2}, v_{N-3}); that at the $+x$ 20% mask the colouring is $\operatorname{sign}(w)\operatorname{sign}(v_2)$ (x does not flip); a finite contrast obstruction on $\mathbb{R}^{24}$ (constant steps never realise the violator occupancy for any remainder; a slow-then-fast block $(0.4^{21}, 2.8^3)$ does, while $(1.0^{21}, 2.0^3)$ does not); and that a coherent $\times 8$ rescaling of the last twelve Jacobi γ's on χ_3 at $w = 9$ *does* produce a v_2-violator, so V_2 is not a theorem of abstract Jacobi-positive chains. A gap-then-spike weight adversary on the real fold-grid at $8\times$ and $64\times$ never produces a V_2-violator and always breaks Jacobi positivity. The hydra-legal reduction is the finite profile lemma V_3': the last fourteen Prüfer steps at the mask lie in the region $\mathcal{A}_{15}$ of sequences that are V_2-regular for *every* incoming remainder $\eta \in [0, \pi)$. That region is strictly smaller than V_2 (finite window, Wronskian/mesh, no product history). On the pin windows the source steps lie in $\mathcal{A}_{15}$ (Path A: 0 remainder-violators; every $(1, 1, 1)$ that occurs is V_2-regular). The chain $T_2' \Leftarrow g_i \geq 3/8 \Leftarrow$ MED-CAP $\Leftarrow X_n \Leftarrow V_2 \Leftarrow V_3'$ stands, with V_3' the remaining named lemma. No RH claim.

Status. Lemmas 1–3 are proved (algebraic identity; finite step-region dichotomy). Lemma 2 is proved (mask geometry). Lemma 4 is a machine-checked negative (an $8\times$ last-12 γ-scale on χ_3 $w = 9$ is a v_2-violator). Lemma 5 is a machine-checked negative (weight-cluster on FRAME-A $w = 9$ never violates, and breaks positivity). Proposition 1 names V_3' and records Path A on the pin windows. The hydra test of [3] is met: V_3' is strictly more elementary than V_2. No RH claim.

1 Recollection

V_2 of [2] §5: if the last three x-order run lengths of $\operatorname{sign}(v_2\, w\, x)$ at the 20% x-mask are $(1, 1, 1)$, then among the four preceding at least two are ≤ 2. The five-run pattern $(\ldots, r, r, 1, 1, 1)$ with $r \geq 3$ is a strictly stronger combinatorial statement (the spike $(2^8, 3, 3, 1, 1, 1)$ is V_2-regular; gate G2) and is not the named remainder. The r365 anatomy [3]: v_2 is the degree-$(N-2)$ arithmetic-comb chain polynomial on the smooth-border nodes (source Jacobi, not mesh-only); w is the explicit smooth-fold arm; a 2-regular factor does not imply V_2; the 134-window surface has 0 violators. The r372 dictionary is a theorem: $\lfloor\theta/\pi\rfloor$ runs of the chain-accumulated Jacobi–Prüfer phase equal the sign-runs of v_2 (181/181, XOR 181/181, interior progress ≥ 2.062). Run lengths *are* phase dwell times. This note attacks V_2 with that dictionary, under the lemma-first charter (exactly one named lemma; exit proved / refuted / reduced).

2 The exact discrete step

Write v_n for the scaled monic chain value of degree $n = N - 2$ and v_{n-1} for the previous scaled value, as produced by the three-term recurrence of [2] §2 (coefficients $(\hat{\alpha}_k, \gamma_{k+1})$, log-scale L_s). Let $\varphi(x) = \operatorname{atan2}(v_n(x), v_{n-1}(x))$ and $r(x) = \sqrt{v_n(x)^2 + v_{n-1}(x)^2}$.

Lemma 1 (Wronskian step)**.** *For any two points x, y at which $r(x)r(y) > 0$,*

$$\sin\bigl(\varphi(y) - \varphi(x)\bigr) \;=\; \frac{v_n(y)\, v_{n-1}(x) - v_{n-1}(y)\, v_n(x)}{r(x)\, r(y)}.$$

Equivalently, writing $\mathbf{u} = (v_n, v_{n-1})$, the identity

$$\bigl(\mathbf{u}(x) \wedge \mathbf{u}(y)\bigr)^2 + \bigl(\mathbf{u}(x) \cdot \mathbf{u}(y)\bigr)^2 \;=\; |\mathbf{u}(x)|^2\, |\mathbf{u}(y)|^2$$

holds in any field of characteristic not 2 (Lagrange).

Proof. The two-dimensional Lagrange identity $(ad - bc)^2 + (ac + bd)^2 = (a^2 + b^2)(c^2 + d^2)$. The sine formula is the imaginary part of $e^{i(\varphi(y)-\varphi(x))} = (\mathbf{u}(y) \cdot \overline{\mathbf{u}(x)})/|\mathbf{u}(x)||\mathbf{u}(y)|$ in the identification $\mathbf{u} \leftrightarrow v_n + iv_{n-1}$. □

The degree-accumulated Prüfer phase θ of the r372 dictionary is the lift of φ along the recurrence in the degree direction, then unwrapped along x. Its spatial steps $\Delta\theta_i = \theta(x_{i+1}) - \theta(x_i)$, reduced to $(-\pi, \pi]$ and oriented so that $\sum \Delta\theta_i > 0$, are the dwell increments: a π-bin change (a v_2 sign-flip) occurs between consecutive nodes iff those nodes occupy distinct $\lfloor\theta/\pi\rfloor$ levels. Lemma 1 identifies the sine of the *principal* step with the scaled Wronskian; the accumulated θ-step agrees with the principal step up to the 2π-unwrap, which the companion script gates on the pin windows.

Lemma 2 ($+x$ mask, x constant)**.** *On the last nine bulk-border nodes at the $+x$ 20% hull mask, $x > 0$. Consequently $\operatorname{sign}(v_2 wx) = \operatorname{sign}(v_2)\operatorname{sign}(w)$ there (the r365 XOR identity, with no x-flip).*

Proof. The mask keeps the central 60% of the hull. The $+x$ edge is at $x \approx +0.60$ (hull near $[-1, 1]$). The last nine bulk nodes sit in a mesh neighbourhood of that edge, on the positive side. The identity is the product rule. □

Thus V_2 at the load-bearing edge is a statement about the XOR of two sign fields, of which v_2 is the Jacobi–Prüfer dwell sequence and w is construction-explicit. On the MAIN family (w a single bulk run) the XOR is vacuous and V_2 is purely a v_2-dwell statement.

3 The contrast obstruction

A run of length r is r consecutive nodes in one π-bin, hence $r - 1$ steps totalling less than π. For $r \geq 3$ the mean interior step is $< \pi/r \leq \pi/3 \approx 1.047$. Three consecutive 1-runs are three consecutive π-crossings: the terminal steps must be large (empirically $\gtrsim 2.2$). The violator $(\ldots, r, r, 1, 1, 1)$ is therefore a *slow-block then fast-block* in the step sequence, not a uniform speed.

Lemma 3 (contrast dichotomy)**.** *Let $d = (\Delta_0, \ldots, \Delta_{23}) \in (0, \pi)^{24}$ be a rigid step sequence, and let $\operatorname{occupancy}(\eta, d)$ be the $\lfloor\theta/\pi\rfloor$ run lengths of the walk $\theta_0 = \eta$, $\theta_{i+1} = \theta_i + \Delta_i$, on 25 nodes.*

1. *Constant steps $\Delta_i \equiv \Delta$ for $\Delta \in \{0.8, 1.0, 1.2, \pi/2, 1.8, 2.0\}$: among 1001 remainders $\eta \in [0, \pi)$, 0 V_2-violators and 0 five-run patterns $(r, r, 1, 1, 1)$.*

2. *Slow-then-fast blocks of the form* (s^{21}, f, f, f). *At* $(s, f) = (0.4, 2.8)$ *there are* $347/1001$ *official* V_2*-violators (and* $674/1001$ *five-run patterns). At* $(s, f) = (1.0, 2.0)$ *both sets are empty.*

A rigid violator, when it occurs, is a slow block adjacent to a fast block, not a constant speed. The two gated profiles separate at a step-ratio between 2 *and* 7.

The companion script re-proves the dichotomy by exact occupancy counting (gate G3–G4). It is a finite statement about $\mathbb{R}^{24}$, independent of the source. The Jacobi coefficients $(\hat{\alpha}_k, \gamma_k)$ do *not* depend on x: there is no "coefficient jump at the mask". The x-variation of the step is entirely the oscillation of the Christoffel/Wronskian profile $K_n(x,x)/r(x)^2$. For a Chebyshev envelope that profile varies by $\sim 6\%$ across nine mask nodes (arcsine at $x = 0.6$); the natural OP oscillation is period ~ 2 mesh steps (interior versus crossing), which produces mixed run lengths $\{1, 2, 3\}$, not a one-sided slow-then-fast block. A Chebyshev θ-compress at $+x$ (19 tails $(1, 1, 1)$ among 108 warp-windows) produces 0 V_2-violators: gradual compression yields mixed prev4, V_2-regular of the same class as χ_3 index 16.

4 Adversaries against the source

Lemma 4 (not a theorem of Jacobi-positive chains)**.** *There exists a Jacobi-positive coefficient perturbation of a real window that realises the* v_2*-violator. Concretely: on* χ_3 *at* $w = 9$ *(*$N = 184$*), multiplying the last twelve* γ_k *by* 8 *produces* v_2*-runs ending* $(3, 1, 3, 3, 1, 1, 1)$*, which fails* V_2*. The same window with scale* 6*, or with last* 4*,* 8*, or* 20 *coefficients scaled by any tested factor in* $\{1.2, \ldots, 12\}$*, does not violate.*

The source last-eight consecutive $|\log(\gamma_{k+1}/\gamma_k)|$ is at most 0.25 (ratio ≤ 1.28), and $\max_k |\gamma_k - \frac{1}{4}| \leq 0.077$ on the pins (last three ≤ 0.041). The adversarial scale 8 sits far outside those bounds. V_2 therefore uses the source, as r365 demanded; it is not abstract product combinatorics (already r365) and not abstract Jacobi-positivity.

Lemma 5 (weight-cluster does not fire)**.** *On FRAME-A* $w = 9$*, depleting* $n_{\text{gap}} = 6$ *arithmetic atoms just before the* $+x$ *mask and boosting* $n_{\text{spk}} = 3$ *atoms at the mask, at boosts* $\{8, 64\}$*, never produces a* v_2 *or colouring violator of* V_2*. Both perturbations break Jacobi positivity (*$\gamma_k < 0$ *for some* $k \leq N - 2$*).*

The natural source-side recipe for a density step (a mass dip then a cluster at the mask) is incompatible with L^* positivity on this window, and even after positivity is lost the occupancy is not the violator (the tail stays a Chebyshev stub of type $(\ldots, 6, 2, 1)$). Arithmetic weight ratios at the mask are already wild (consecutive ratios up to 25–70 on the pins); the OP zeros are more regular than the atoms.

5 The smaller lemma V_3'

Definition 1 (V_3')**.** *Let* $L = 15$ *and let* $(\Delta\theta_i)_{i=0}^{L-2}$ *be the spatial steps of the degree-accumulated Jacobi–Prüfer phase of* v_2 *on the last* L *bulk-border nodes at the* $+x$ 20% *mask, each reduced to* $(-\pi, \pi]$ *and oriented so that* $\sum \Delta\theta_i > 0$*. Write* $\mathcal{A}_L \subset \mathbb{R}^{L-1}$ *for the region of sequences such that for every incoming remainder* $\eta \in [0, \pi)$*, the occupancy of the walk* $\theta_0 = \eta$*,* $\theta_{i+1} = \theta_i + \Delta\theta_i$ *is* V_2*-regular: if it ends* $(1, 1, 1)$*, then among the four preceding runs at least two are* ≤ 2*. Condition* V_3' *is: the source* 14*-tuple lies in* $\mathcal{A}_{15}$*.*

Proposition 1 (reduction)**.** V_3' *is strictly more elementary than* V_2 *in the sense of the hydra ward of [3] §5:*

1. Finite. *A point of* $\mathbb{R}^{14}$ *and a compact scan of* $\eta \in [0, \pi)$*.* V_2 *is an implication over a variable-length run history of an* $O(N)$*-cell product colouring.*

2. Closer to the construction. *The coordinates are the Wronskian steps of the Jacobi chain on the mask mesh (Lemma 1); no pairing, no* w*, no MED-CAP. The* 15 *nodes are the tent/mask geometry.*

3. Fewer quantifiers. $\forall\eta$ *on a compact interval, of a* 14*-tuple, versus four preceding runs of unbounded length in the product* $\operatorname{sign}(v_2 w x)$*.*

V_3' *implies* V_2 *for* v_2*-runs: the actual remainder is one* η*. At the* $+x$ *mask, Lemma 2 reduces the colouring to* $\operatorname{sign}(w)\operatorname{sign}(v_2)$*. On MAIN (*$w$ *constant on the bulk) this is* v_2*-only. On a* w*-burst, Chebyshev times Nyquist never violates (r365 Lemma 6, re-gated). On the pin windows every* $(1, 1, 1)$ *produced by an* $\mathcal{A}_{15}$*-profile under remainder variation is* V_2*-regular; on* χ_3 *index* 16 *the actual occupancy is* prev4 $= (3, 2, 1, 3)$ *and the* η*-scan has two regular* prev4 *types. The chain is*

$$T_2' \impliedby g_i \geq \tfrac{3}{8} \impliedby \text{MED-CAP} \impliedby X_n \impliedby V_2 \impliedby V_3'.$$

V_3' is logically stronger than V_2 for v_2-runs (all remainders, not just the actual one). That is the content of the reduction: the occupancy is an alignment, the obstruction is the speed profile. Lemma 3 identifies $\mathbb{R}^{14} \setminus \mathcal{A}_{15}$ with slow-then-fast contrast blocks. The remaining lemma is that the source Wronskian profile at the mask has no such block.

Proposition 2 (Path A on the pins)**.** *On the six pin windows (four* $w = 9$ *worlds,* χ_4 *index* 53*,* χ_3 *index* 16*), and on four extra FRAME-A rungs (*$kz \in \{15, 21, 25, 33\}$*):*

1. *the last-*15 *Prüfer-step tuple lies in* $\mathcal{A}_{15}$ *(*0 *remainder-violators among* 4001 *remainders at* $L \in \{9, 13, 17\}$*);*

2. *every remainder that produces a tail* $(1, 1, 1)$ *does so with mixed* prev4 $= (3, 2, 1, 3)$ *(the* χ_3 *index-*16 *type), never with two preceding runs* ≥ 3*;*

3. *colouring* V_2 *holds (regression of [3]);*

4. χ_3 *index* 16 *is the* v_2*-triple, even run count,* $n_0 = 2$*;* χ_4 *index* 53 *is the* w*-driven triple, odd, saturating prefix* $(1, 2, 6, 5, 4, 4)$*.*

This is a pin census of V_3'*, not a cofinal law.*

The r365 hydra candidates (mesh-only v_2; 2-regular v_2; w-only; Freud / $\gamma_k \to 1/4$) fail the elementary test or are false. V_3' does not: it is a finite region in the Wronskian coordinates of a 15-node mask window. It does not reduce toward L* (no coefficient-asymptotic is assumed; Lemma 4 shows that a large coherent γ-scale, far outside the source, *does* leave $\mathcal{A}_{15}$).

6 What remains

To prove V_3' cofinally: show that the source Wronskian profile $K_n(x, x)/r(x)^2$ on the last 15 bulk nodes has no slow-then-fast block of the Lemma 3 type. The envelope of a near-Chebyshev Jacobi chain varies too slowly at $x \approx 0.6$ (relative change $O(1/N)$ across the window); the OP oscillation is two-step, not a one-sided burst. The coefficient bounds of the source ($|\gamma - 1/4| \leq 0.077$ on the pins, consecutive ratio ≤ 1.28) exclude the only explicit coefficient adversary that left $\mathcal{A}_{15}$ (Lemma 4). Making that comparison a theorem — a quantitative Sturm bound on $K_n(\,\cdot\,, \cdot\,)$ at the mask, from the tent/mesh and the von Mangoldt weights — is the remaining work. It is a statement about fifteen nodes and the Wronskian, not about V_2.

T1 / C_K unchanged. The cubic-mass bound of [1] still carries C_K. No RH claim.

7 Machine checks

Every numbered lemma is re-proved in `verify_v2_lemma.py` (same directory). Standalone: Lagrange identity over $\mathbb{Q}$; contrast dichotomy; Chebyshev mask and θ-compress; the named combinatorial violator. Construction: Wronskian residual, Path A on the pins, γ bounds, the $8\times$ last-12 adversary, $+x$ sign, the r372 dictionary on the pins, colouring V_2 regression, the weight-cluster pair at boosts 8 and 64. Exit line: `V2 LEMMA V3 VERIFIED`.

Claim boundary

Finite identities on an explicit Wronskian, a finite contrast dichotomy in $\mathbb{R}^{24}$, a named coefficient adversary, a weight-cluster negative, and a strictly smaller finite profile lemma V_3' to which V_2 reduces. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

[1] S. Hamann, *A median cap for gaps of a tiled integer packing*, standalone note, August 2026, `rh/problem/medcap_lemma.tex`.

[2] S. Hamann, *The length invariant X_n of a folded sign field*, standalone note, August 2026, `rh/problem/xn_invariant.tex`.

[3] S. Hamann, *Run-length regularity V_2 at the x-mask*, standalone note, August 2026, `rh/problem/v2_regularity.tex`.